

DSA / Algorithms (Abdul Bari) - Day 30

45-day plan mapped to the Algorithms playlist

Traveling Salesman (DP)

TSP via dynamic programming.

Watch these Abdul Bari videos

> 4.7 [New] Traveling Salesman Problem - Dynamic Programming using Formula

Key concepts

- Visit all cities once, return to start, min cost.
- DP over subsets (bitmask).
- $O(2^n * n^2)$.

Key idea

$g(i, S) = \min \text{ over } j \text{ of } \text{cost}(i, j) + g(j, S - \{j\})$.

Tasks (1.5h)

- ☐ Watch the video
- ☐ Understand the subset DP
- ☐ Trace a 4-city example

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0